

Helium gas or liquid nitrogen for the first thermal shield of a 4 K cryostat: conjugate heat-transfer budget, coolant operating maps and work ledger

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Abstract

The first thermal shield of a 4 K cryostat intercepts most of the heat arriving from the 300 K environment. It can be held near 50 K by circulated helium gas, as in fusion and accelerator cryostats, or at 77 K by liquid nitrogen, as in magnet cryostats. This paper separates that choice into three questions and answers each with a dedicated model. (1) The passive penalty of the warmer shield is computed with a three-dimensional multi-region conjugate heat-transfer model of a representative cryostat section that resolves view-factor radiation and the conduction of a 65-line coaxial harness; the view-factor matrix is shown to require row normalisation in a low-emissivity enclosure, after which the model reproduces the closed-form loads to 4 %. (2) The two coolants are compared in a channelled cold plate at a matched 9.4 W duty over Reynolds numbers 500–10 000, laminar to turbulent, and helium pressures of 1–18 bar, with properties from the National Institute of Standards and Technology (NIST) reference equations of state, three-grid verification for both fluids and energy closure below 0.2 %. (3) The routes are priced in room-temperature power with ideal Carnot multipliers and with published hardware efficiencies. The warmer shield costs nothing at the first stage, 8.3 against 8.4 W; its penalty is 106 mW at the 4 K stage, 86 % of it harness conduction, so it is a property of the wiring rather than of the cryogen. In the cold plate nitrogen gives five times helium’s heat-transfer coefficient in laminar flow and six times in turbulent flow, and holds the plate within 4 K of its inlet where helium at 1 bar floats 21 K above and exceeds 77 K below a

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Reynolds number of about 1400; at 18 bar helium's pumping power falls 8.5 times below nitrogen's with heat transfer unchanged, so the ranking depends on the basis and the operating point. The ideal ledger favours nitrogen by 21 % for a bare shield and helium by 8 % once multi-layer insulation removes most of the first-stage load; with hardware efficiencies helium is cheaper by 35–50 % in both cases, and nitrogen becomes competitive only if the harness conduction to 4 K is cut by about a factor of three. Operating maps, uncertainty bands and the complete case set are provided.

Keywords: cryostat, thermal shield, conjugate heat transfer, liquid nitrogen, helium gas, pulse-tube cryocooler, exergy, OpenFOAM

Nomenclature

Roman symbols

A	area [m ²]
c_p	specific heat [J kg ⁻¹ K ⁻¹]
D_h	hydraulic diameter [m]
f	Darcy friction factor [-]
F_{ij}	view factor from face i to face j [-]
h	heat-transfer coefficient [W m ⁻² K ⁻¹]
J	radiosity [W m ⁻²]
k	thermal conductivity [W m ⁻¹ K ⁻¹]
$\dot{m}$	mass flow rate [kg s ⁻¹]
Mo	Mouromtseff number [-]
N	number of MLI layers [-]
NTU	number of transfer units [-]
Nu	Nusselt number hD_h/k [-]
p	pressure [Pa]
Pr	Prandtl number [-]
$\dot{Q}$	heat load [W]
q	heat flux [W m ⁻²]
Re	Reynolds number $\rho U D_h / \mu$ [-]
$\dot{S}_{\text{gen}}$	entropy generation rate [W K ⁻¹]
T	temperature [K]
T_0	heat-rejection temperature, 300 K
U	mean velocity [m s ⁻¹]
$\dot{W}$	wall power [W]
$\dot{W}_p$	pumping power [W]
y^+	wall coordinate [-]

Greek symbols

β	thermal expansion coefficient [K ⁻¹]
ε	emissivity; HX effectiveness [-]
η_i	stage efficiency, fraction of Carnot [-]
κ_{eff}	effective harness conductivity [W m ⁻¹ K ⁻¹]
μ	dynamic viscosity [Pa s]
ρ	density [kg m ⁻³]
σ	Stefan–Boltzmann constant

Subscripts 0 ambient; 1 first stage; 2 second (4 K) stage; b bulk; c conduction; r radiation; sat saturation; w wall.

Acronyms CFD computational fluid dynamics; CHT conjugate heat transfer; DTT Divertor Tokamak Test facility; GCI grid convergence index; ITER International Thermonuclear Experimental Reactor; LHC Large Hadron Collider; LHD Large Helical Device; MLI multi-layer insulation; NIST National Institute of Standards and Technology; NTU number of transfer units; OVC outer vacuum can; PTFE polytetrafluoroethylene; PTR pulse-tube refrigerator; SIMPLE semi-implicit method for pressure-linked equations; SST shear-stress transport; STL stereolithography (surface file format).

1. Introduction

A cryostat that holds a payload at 4 K places at least one thermal shield between the payload and the 300 K vacuum vessel. The shield intercepts the radiation from the vessel and the conduction along supports and wiring, and it does so at a temperature where refrigeration is one to two orders of magnitude cheaper than at 4 K [1, 2, 3]. Two ways of cooling this first shield are in wide use. Large superconducting systems circulate pressurised helium gas through cooling tubes on the shield: the thermal shield of the International Thermonuclear Experimental Reactor (ITER) runs on helium at 80 K and 1.8 MPa [4], the shield of the Divertor Tokamak Test facility (DTT) on the same loop conditions [5], the shields of the Large Helical Device (LHD) on gaseous helium in parallel paths [6], and the shields of the Large Hadron Collider (LHC) on helium at 50–75 K [7, 8]. Magnet and detector cryostats, and the older generation of helium cryostats, use a liquid-nitrogen bath or loop at 77 K [9, 10]; Li et al. made the practical case for it three decades ago, that a boiling cryogen holds the shield temperature constant and carries a stored reserve, whereas a helium-gas shield warms as soon as its refrigerator stops [11]. Compact laboratory cryostats, including the dry dilution refrigerators that host superconducting quantum processors, use the first stage of a helium pulse-tube refrigerator (PTR) at 40–50 K instead [12, 13, 14]. As these laboratory systems grow toward the multi-metre cryostats now being built for quantum platforms [15, 16], they inherit the shield-cooling question that fusion and accelerator cryostats answered with circulated fluids, and the cryogenic plant becomes a dominant part of the system’s energy cost [17, 18, 19].

The published treatments of this choice fall into four strands that do not meet. First, shield design temperatures are optimised with lumped static budgets and refrigeration-power minimisation: Lagrange-multiplier optimisation of nitrogen, neon and helium-gas intercepts on supports and in insulation [11], multi-stage shields for transmission-line cryostats [20], fusion thermal shields [21, 5], and single-shield helium cryostats brought to zero boil-off by load minimisation [22]. These are one-dimensional: they return a stage total, not where the load lands. Second, wiring loads are measured and modelled per cable [14, 23, 24], again as conduction integrals that assign the harness a single resistance and leave it uncoupled from the radiation field. Third, surface-to-surface radiation in computational fluid dynamics (CFD) codes has been validated on enclosure benchmarks and applied to fusion

vacuum vessels [25], but not coupled to temperature-dependent harness conduction in a cryostat shield, and the behaviour of agglomerated view-factor matrices in such enclosures has not been examined. Fourth, cryogenic heat exchangers are optimised by entropy-generation minimisation [26, 27, 28] and liquid-nitrogen cooled components are now modelled with combined analytical and CFD frameworks [29], yet helium gas and liquid nitrogen have not been ranked against each other as forced-convection first-shield coolants on a common duty across flow regimes and pressures, with heat-transfer, hydraulic, effectiveness and second-law metrics reported together and the single-phase margin of the nitrogen verified. Finally, the cryocooler and liquefier efficiencies that separate an ideal Carnot ledger from a real one are well documented [30, 8, 31], but the two ledgers have not been reconciled for this choice.

The gap is therefore not a missing number but a missing junction. What is absent is a treatment that (i) resolves shield radiation and harness conduction together in three dimensions on one geometry, (ii) compares the two coolants like for like in the cold plate that geometry implies, and (iii) prices both routes with ideal and with hardware efficiencies, so that the conditions under which each route wins can be stated rather than assumed.

This paper therefore asks three questions about a representative 4 K cryostat section with a 65-line coaxial harness (Figs. 1 and 2):

Q1 (passive penalty). What does raising the first shield from 50 K to 77 K cost in heat load at the first stage and at the 4 K stage, and which mechanism carries the cost?

Q2 (coolant performance). If the first-stage duty is removed by a coolant circulated through a channelled cold plate, how do helium gas and liquid nitrogen compare across flow regimes, pressures and bases of comparison, and where is each fluid’s best operating point?

Q3 (work ledger). What do the two routes cost in room-temperature power, with ideal Carnot multipliers and with the efficiencies of real hardware?

Q1 is answered with a three-dimensional multi-region conjugate heat-transfer (CHT) model that resolves view-factor radiation and temperature-dependent harness conduction. Its stage-integrated loads are verified against closed-form budgets; its contribution is the spatial distribution of the load and the split between harness families. Q2 is answered with a grid-converged cold-plate model run for both fluids over $Re = 500\text{--}10^4$ and helium pressures of

1–18 bar, with properties from the NIST reference equations of state [32]. Q3 combines the stage loads of Q1 with Carnot multipliers and with published cryocooler and liquid-nitrogen plant efficiencies [30, 33, 31]. The dilution unit or any payload below 4 K is outside the scope; the results apply to any 4 K cryostat with a wired first shield.

2. Models and methods

2.1. Governing equations

Both models solve steady conservation equations with OpenFOAM v2412 [34]. In the enclosure model the inter-stage gaps are vacuum, so each gap region carries only the radiative exchange and each solid region carries conduction with a temperature-dependent conductivity,

$$\nabla \cdot [\kappa_{\text{eff}}(T) \nabla T] = 0 \quad \text{in the harness bundles,} \quad (1)$$

coupled at the bundle surfaces to the radiative flux of the enclosing gap. Radiation is solved as a grey-diffuse radiosity system on the agglomerated boundary faces,

$$J_i = \varepsilon_i \sigma T_i^4 + (1 - \varepsilon_i) \sum_j F_{ij} J_j, \quad q_{r,i} = J_i - \sum_j F_{ij} J_j, \quad (2)$$

where F_{ij} are geometric view factors computed by ray tracing between agglomerated faces (`viewFactor` model). For two parallel grey surfaces Eq. (2) reduces to the closed form $q_r = \sigma(T_{\text{hot}}^4 - T_{\text{cold}}^4)/(1/\varepsilon_{\text{hot}} + 1/\varepsilon_{\text{cold}} - 1)$ [35], which is used as the verification anchor.

The cold-plate model solves the steady Reynolds-averaged (or, in the laminar range, the plain) continuity, momentum and energy equations for a single-phase coolant,

$$\nabla \cdot (\rho \mathbf{U}) = 0, \quad (3)$$

$$\nabla \cdot (\rho \mathbf{U} \mathbf{U}) = -\nabla p + \nabla \cdot [(\mu + \mu_t)(\nabla \mathbf{U} + \nabla \mathbf{U}^T)] + \rho \mathbf{g}, \quad (4)$$

$$\nabla \cdot (\rho \mathbf{U} h) = \nabla \cdot \left[\left(\frac{k}{c_p} + \frac{\mu_t}{Pr_t} \right) \nabla h \right], \quad (5)$$

with $\mu_t = 0$ for laminar runs and μ_t from the k - ω shear-stress transport (SST) model [36] for $Re \geq 5000$ ($Pr_t = 0.85$). The sweep is run as pure forced convection ($\mathbf{g} = 0$) so that the ranking does not depend on the orientation

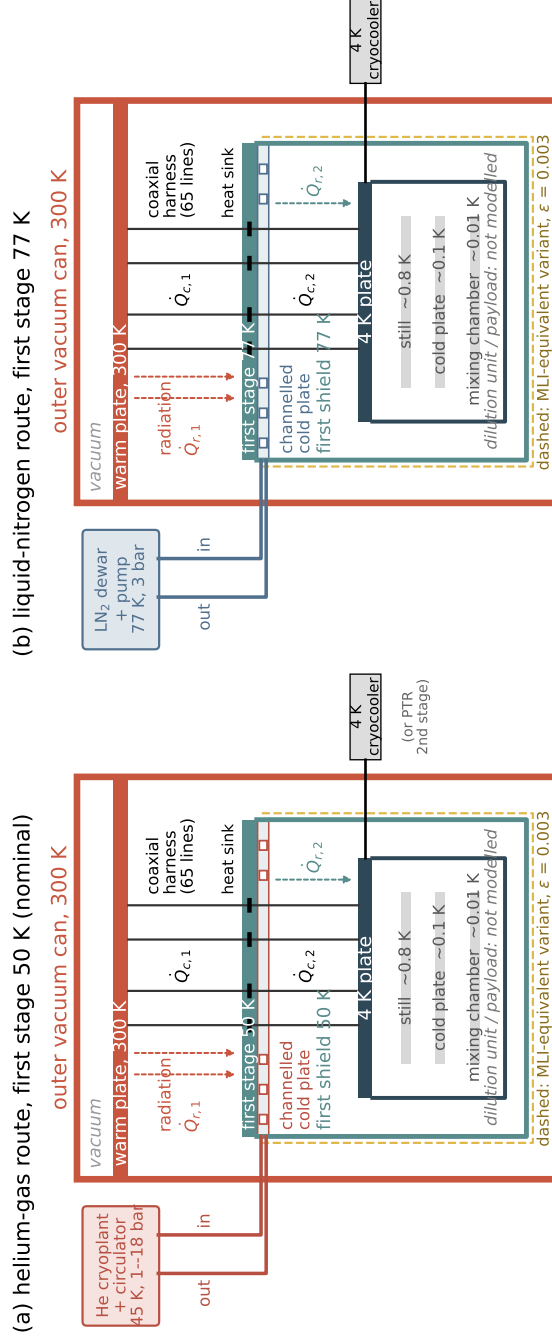


Figure 1: Cross-section of the cryostat class studied, with (a) the helium-gas route and (b) the liquid-nitrogen route for the first shield. The dashed region from the 300 K plate to the 4 K plate is the enclosure CHT model and the channelled first-stage plate is the cold-plate model; the greyed stages below 4 K are shown for context only and are not modelled.

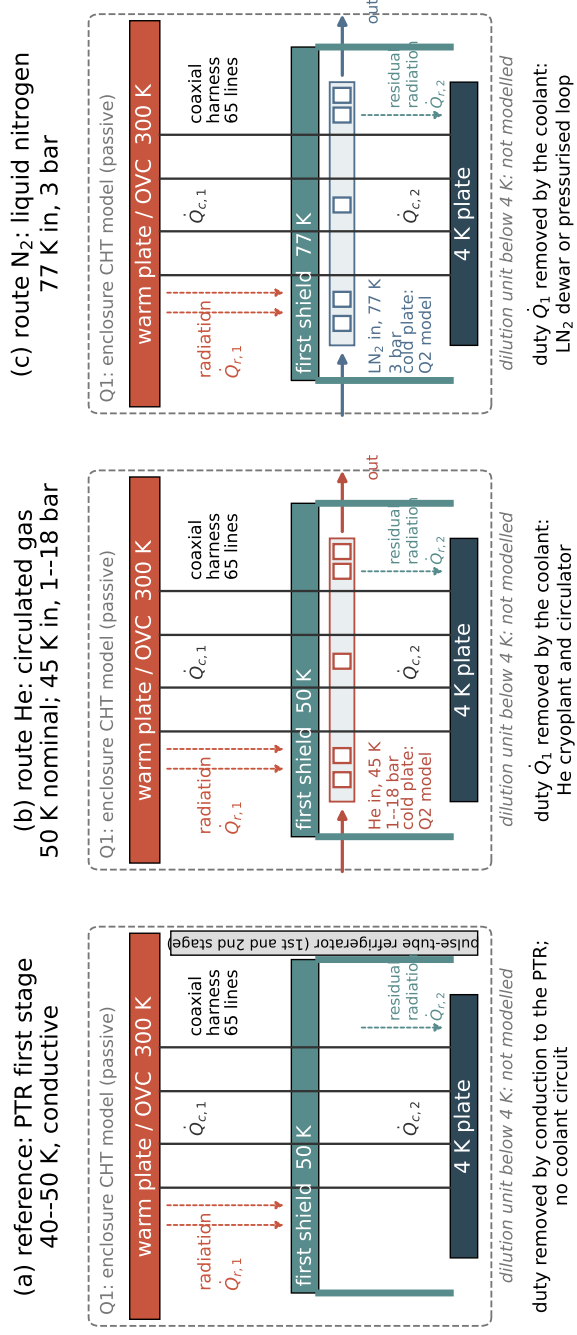


Figure 2: The three first-shield cooling routes and the two models. (a) Reference architecture with a conductively coupled PTR first stage; (b) circulated helium gas through a channelled cold plate; (c) liquid nitrogen through the same cold plate. The dashed box is the domain of the passive enclosure CHT model (Q1); the dotted box is the domain of the active cold-plate model (Q2). The loads $\dot{Q}_1 = \dot{Q}_{r,1} + \dot{Q}_{c,1}$ and $\dot{Q}_2 = \dot{Q}_{r,2} + \dot{Q}_{c,2}$ enter the work ledger ($Q3$), $\dot{W} = \sum_i \dot{Q}_i (T_0 - T_i) / (T_i \eta_i)$, with $\eta_i = 1$ (ideal) or the measured fraction of Carnot (real).

of the plate; the buoyancy parameter Gr/Re^2 is reported for every run, and three runs with gravity acting along the heated wall quantify the mixed-convection sensitivity (Section 4). Helium is treated as an ideal gas with variable density, $\rho = pM/(R_u T)$, because its density changes by 5–26 % between inlet and outlet and by up to a factor of 2 between bulk and heated wall; liquid nitrogen is treated with the Boussinesq approximation, $\rho = \rho_0[1 - \beta(T - T_0)]$, valid because $\beta\Delta T < 0.03$ in every run.

2.2. Enclosure CHT model (Q1)

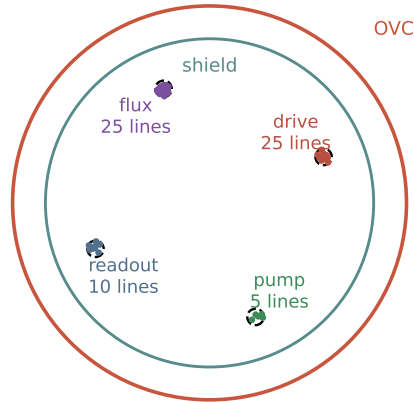
The domain is a $0.718 \times 0.718 \times 0.335$ m section of a cryostat between the warm plate and the first-stage cold plate (Fig. 4; Table 1). Two vacuum gap regions (outer: outer vacuum can (OVC) to shield; inner: warm plate to cold plate and shield) and four solid cylinders of 20 mm diameter spanning the 0.335 m height are meshed with a $30 \times 30 \times 14$ background grid refined to level 2 around the cylinders ($0.2\text{--}0.4 \times 10^6$ cells in total). Each cylinder lumps one functional family of a cabled 65-line control harness [14]: 25 drive, 25 flux, 10 readout and 5 pump lines, all of the UT-085 stainless-steel/PTFE coaxial construction. The family is represented by an effective conductivity

$$\kappa_{\text{eff}}(T) = \frac{n_k}{A_{\text{model}}} \sum_i A_i k_i(T), \quad (6)$$

where n_k is the number of lines in the family, A_i the cross-section of cable component i (outer conductor 1.585 mm^2 and centre conductor 0.205 mm^2 of 304 stainless steel, polytetrafluoroethylene (PTFE) dielectric 2.001 mm^2), $k_i(T)$ the NIST cryogenic conductivity fits, and A_{model} the effective conduction area of the modelled cylinder, calibrated once per geometry so that the resolved conduction reproduces the harness integral $n_k L^{-1} \sum_i A_i \int k_i dT$ (Supplementary Table S4 lists the calibrated areas and the fifth-order polynomial coefficients of κ_{eff} for every bundle and stage span; fit error below 0.5 %; Fig. 3).

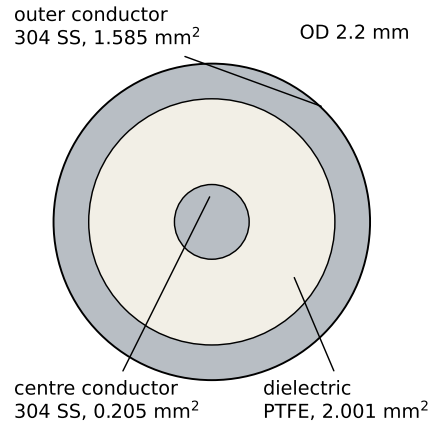
Verification and uncertainty. Stage-integrated radiation is compared with the parallel-plate closed form and harness conduction with the analytic integral; both agree within the tolerances reported in Section 3. Because the view-factor matrix is built on agglomerated faces, the radiative load was re-computed at 250, 400 and 800 coarse faces per patch and its closure $\sum_j F_{ij}$ monitored (Supplementary Table S5). The emissivity of polished cryogenic surfaces is uncertain by roughly a factor of 1.5 [37]; the $\pm 50\%$ runs give the

(a) 65 lines in four families



each family $\rightarrow$ one 20 mm cylinder
at $r = 0.15$ m (plan view, to scale)

(b) UT-085 cable cross-section



$$(c) \kappa_{eff}(T) = \frac{n_k}{A_{model}} \sum_i A_i k_i(T)$$

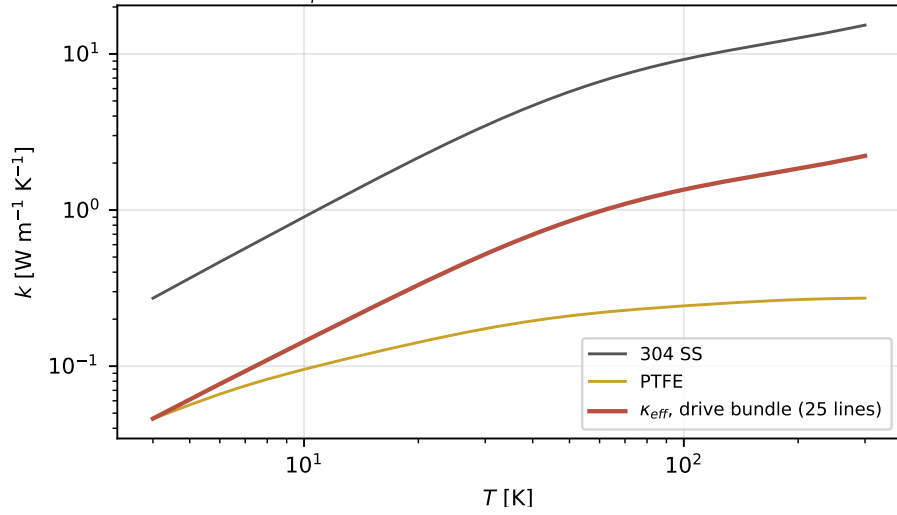


Figure 3: Harness representation: (a) the 65 coaxial lines in four functional families, each lumped into one 20 mm cylinder; (b) the UT-085 cable cross-section with the component areas of Eq. (6); (c) the component conductivities and the resulting effective conductivity of the drive bundle.

Table 1: Enclosure model: geometry, boundary conditions and numerical settings.

Domain	$0.718 \times 0.718 \times 0.335$ m; OVC and shield as snapped stereolithography (STL) surfaces; four coax cylinders $\varnothing 20$ mm at radius 0.15 m
Regions	domain0 (OVC–shield gap), domain1 (warm plate–cold plate/shield gap), coax.L1..L4 (solids)
Gas in gaps	vacuum: $p = 10^{-3}$ Pa, $g = 0$, $\mu = 10^{-11}$ Pa s (no convection, negligible gas conduction; free-molecular bound $< 1\%$ of the stage load, Supplementary S1)
Temperatures	fixed: warm plate and OVC at T_{hot} ; shield and cold plate at T_{cold} ; cases (300, 50), (300, 77), (50, 4), (77, 4) K
Emissivity (baseline)	warm surfaces 0.04 (polished Al); shield and cold plate 0.02 (polished Cu-plated steel); harness 0.02
Emissivity (variants)	$\pm 50\%$ (0.02/0.01 and 0.06/0.03); multi-layer-insulation (MLI) equivalent 0.003 on all first-stage surfaces
Radiation	viewFactor , grey diffuse, one band, solved every 10 iterations; agglomeration 250 faces per patch at the coarsest level (verification at 400 and 800)
Harness contact	bundles fixed at T_{hot} at the warm cap and T_{cold} at the cold cap; no contact resistance; lateral surfaces exchange by radiation only
Solver	chtMultiRegionSimpleFoam , semi-implicit method for pressure-linked equations (SIMPLE), residuals $< 10^{-7}$, 2900 iterations from a uniform initial field

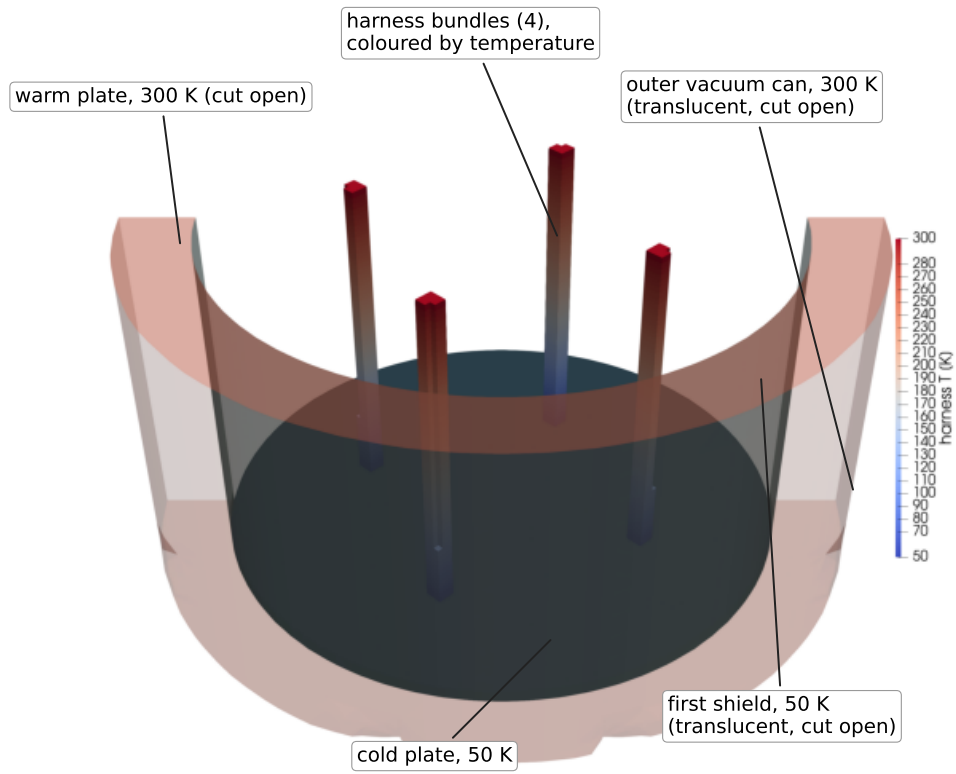


Figure 4: Cut-away of the enclosure CHT domain for the $300 \rightarrow 50$ K case. The warm plate, outer vacuum can and first shield are cut open on the near side; the four harness bundles, coloured by their computed temperature, span the gap from the warm plate to the cold plate. The vacuum gaps between the surfaces carry radiation only.

131 uncertainty band carried through the ledger. An MLI-equivalent variant with
 132 $\varepsilon = 0.003$ on the first-stage surfaces represents a 10–20 layer blanket [38, 1].

133 *What the three-dimensional model adds..* For a parallel-plate enclosure with
 134 fixed wall temperatures, the stage-integrated radiative and conductive loads
 135 are analytic, and the CHT model must reproduce them; that agreement is the
 136 model’s verification, not its result. The model contributes what the closed
 137 forms cannot: the distribution of radiative flux over the shield (which sets
 138 local shield temperatures once a real, finitely conductive shield is considered),
 139 the partition of conduction among harness families, the coupling between the
 140 harness and the surrounding radiation field, and, through the same geometry,
 141 the boundary conditions passed to the cold-plate model.

142 2.3. Cold-plate heat-exchanger model (Q2)

143 *Basis of comparison..* The two fluids differ in phase, temperature, density
 144 by a factor of 800 and thermal conductivity by a factor of 3.1, so no sin-
 145 gle common basis is neutral. Three bases are therefore reported: matched
 146 Reynolds number (same flow regime), matched mass flow, and matched
 147 pumping power. All runs remove the same duty of 9.4 W, the conserva-
 148 tive first-stage load of Section 3. Helium enters at 45 K, at 1, 5 or 18 bar (the
 149 ITER shield loop runs at 18 bar [4]); nitrogen enters as a liquid at 77 K in a
 150 3 bar loop, for which $T_{\text{sat}} = 87.9$ K, and the saturation margin $T_{\text{sat}} - T_{\text{w,max}}$
 151 is reported for every run to confirm single-phase operation. Table 2 lists the
 152 operating quantities and Fig. 5 the geometry.

153 *Property correction..* An earlier version of this comparison used helium trans-
 154 port properties that correspond to about 15 K, namely $\mu = 2.7 \mu\text{Pa s}$ and
 155 $k = 0.019 \text{ W m}^{-1} \text{ K}^{-1}$, which understated the helium heat-transfer coefficient
 156 by a factor of about 2.5. All helium results here use the 50 K values of
 157 Table 2.

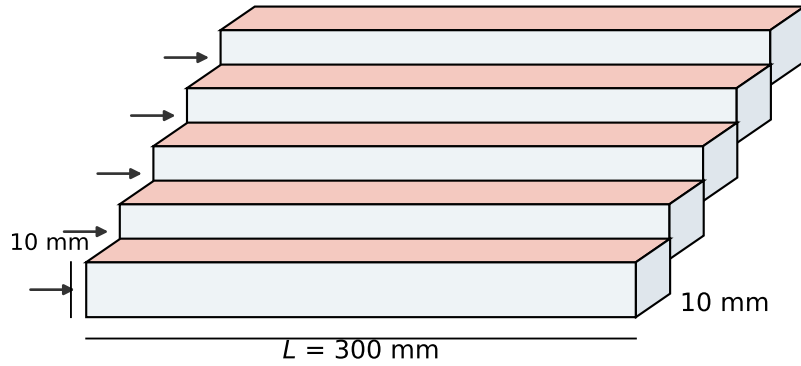
158 *Metrics..* The heat-transfer coefficient is $h = \dot{Q}/[A_{\text{heated}}(T_{\text{w}} - T_{\text{b}})]$ with T_{w}
 159 the area-averaged heated-wall temperature and T_{b} the mean of the inlet and
 160 the mass-flux-weighted outlet temperature; $Nu = hD_{\text{h}}/k$; the Darcy friction
 161 factor $f = 2\Delta p D_{\text{h}}/(\rho U^2 L)$; pumping power $\dot{W}_{\text{p}} = \dot{m}\Delta p/\rho$; $NTU = hA/(\dot{m}c_p)$
 162 and $\varepsilon = 1 - e^{-NTU}$ [39]; the Mouromtseff number $Mo = k^{0.6}\rho^{0.8}c_p^{0.4}\mu^{-0.4}$ [40];
 163 and the exergy destruction $T_0\dot{S}_{\text{gen}}$ from volume integrals of the thermal and
 164 viscous entropy sources [28]. Laminar results are checked against the fully

Table 2: Cold-plate model: geometry, coolant states, properties and numerics. Properties from the NIST reference equations of state via CoolProp [32], evaluated at the inlet state and held constant except for the helium density (ideal gas) and the nitrogen buoyancy term.

Shared geometry and duty		
Channels	five parallel square ducts, one wall heated	
Dimensions	$D_h = 10 \text{ mm}$, $L = 0.30 \text{ m}$; $A_{\text{heated}} = 0.015 \text{ m}^2$, $A_{\text{CS}} = 5 \times 10^{-4} \text{ m}^2$	
Duty	9.4 W, uniform flux $q_w = 627 \text{ W m}^{-2}$ on the heated wall	
Mesh	$180 \times 18 \times 18$ cells per duct, wall-graded (ratio 6); 2.92×10^5 cells; GCI on 86k / 292k / 984k	
Gravity	off (forced convection); on for three sensitivity runs	
	Helium gas	Liquid nitrogen
<i>Coolant state</i>		
Inlet temperature	45 K	77 K
Loop pressure	1, 5 or 18 bar	3 bar ($T_{\text{sat}} = 87.9 \text{ K}$)
Phase	ideal gas, variable density	single-phase liquid (margin reported)
<i>Properties at the inlet state</i>		
Density ρ [kg m^{-3}]	0.961 / 4.80 / 17.3 at 1 / 5 / 18 bar	808
c_p [$\text{J kg}^{-1} \text{K}^{-1}$]	5201	2038
Viscosity μ [$\mu \text{Pa s}$]	6.36	163.5
k [$\text{W m}^{-1} \text{K}^{-1}$]	0.0467	0.1457
Prandtl number Pr	0.71	2.29
β [K^{-1}]	$1/T$ (ideal gas)	5.6×10^{-3}
<i>Flow range</i>		
Reynolds numbers	500, 1000, 1500, 2300 (laminar); 5000, 10 000 (k - ω SST)	
$\dot{m}$ at $Re = 2300$ [kg s^{-1}]	7.31×10^{-4}	1.88×10^{-2}
U at $Re = 2300$ [m s^{-1}]	1.52 (1 bar); 0.084 (18 bar)	0.047
<i>Numerics</i>		
Solver	buoyantSimpleFoam (ideal gas)	buoyantBoussinesqSimpleFoam
Convergence	residuals $< 10^{-6}$	residuals $< 10^{-7}$
Energy closure	per run; $< 0.2 \%$ for every forced-convection case	

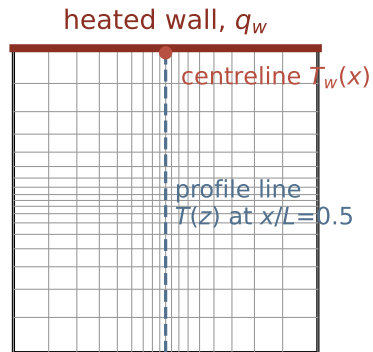
(a) cold-plate heat-exchanger domain

heated wall (top of every duct): uniform $q_w = \dot{Q}/A_{heated} = 627 \text{ W m}^{-2}$



inlet: T_{in} , $\dot{m}$ (He: mass-flow BC; LN₂: velocity BC); outlet: fixed p
 other walls adiabatic and no slip; five square ducts, $D_h = 10 \text{ mm}$, pitch 18 mm
 heated area 0.015 m^2 (top walls), flow area $5 \times 10^{-4} \text{ m}^2$

(b) duct cross-section: mesh and sampling lines



18×18 cells per duct
 two-sided grading, ratio 6
 first cell 0.19 mm
 180 cells along L
 2.9×10^5 cells in total

Figure 5: Cold-plate heat-exchanger model: (a) the five heated square ducts with boundary conditions and dimensions; (b) the graded duct cross-section with the two sampling lines used for the wall-temperature profiles.

165 developed square-duct limits $Nu_\infty = 3.61$ and $fRe = 56.9$ [41] and turbu-
 166 lent results against the Gnielinski correlation [42]. Wall temperature profiles
 167 along the heated-wall centreline are extracted for every run.

168 *Grid convergence..* Three systematically refined meshes (refinement ratio
 169 1.5) were run for both fluids at $Re = 2300$ and the discretisation uncertainty
 170 expressed as a GCI [43] (Section 4). For the turbulent runs the first-cell
 171 y^+ on the graded mesh averages 2.5–5.6 with a maximum of 9.5, within the
 172 range handled by the low-Reynolds wall treatment of the SST model.

173 2.4. Work ledger (Q3)

174 A load $\dot{Q}_i$ absorbed at T_i and rejected at $T_0 = 300$ K costs

$$\dot{W}_i = \frac{\dot{Q}_i}{\eta_i} \frac{T_0 - T_i}{T_i}, \quad (7)$$

175 where η_i is the stage efficiency as a fraction of Carnot. The ideal ledger sets
 176 $\eta_i = 1$: the multiplier is 5.0 at 50 K, 2.9 at 77 K and 74 at 4 K. The real ledger
 177 uses published hardware. A commercial two-stage PTR (Cryomech PT415)
 178 delivers 1.5 W at 4.2 K together with 40 W at 45 K for 10.7 kW of input [33];
 179 allocating the input in proportion to the Carnot work of each stage gives
 180 $\eta_2 \approx 0.011$ at 4.2 K and $\eta_1 \approx 0.045$ at 45 K (Supplementary S3), consistent
 181 with the 1 % and 10–15 % of Carnot that Radebaugh reports for 4 K and
 182 80 K coolers [30]. Liquid nitrogen carries 199 kJ kg⁻¹ of latent heat at 1 bar;
 183 its ideal production work is 0.213 kWh kg⁻¹ [2] and real air-separation plants
 184 need 0.5–0.7 kWh kg⁻¹ [31], an exergy efficiency $\eta_{N_2} \approx 0.3$ –0.4. The 4 K stage
 185 of the nitrogen route is still served by a cryocooler, so its η_2 is the same in
 186 both routes. The ranges above are propagated as bands.

187 3. Results Q1: the passive penalty of a 77 K shield

188 3.1. Radiation verification and the view-factor closure

189 The first-stage radiative load computed with the view-factor matrix as
 190 generated does not converge under agglomeration refinement: 9.46, 8.41, 6.49
 191 and 3.26 W at 60, 250, 400 and 800 coarse faces per patch, against 7.07 W
 192 from the parallel-plate closed form (Fig. 6). The cause is closure. In a closed
 193 enclosure every row of the view-factor matrix must sum to one; the generated
 194 rows sum to between 0.50 and 1.13 (mean 0.97–1.01), and finer agglomeration
 195 worsens the scatter. With surface emissivities of 0.02–0.04 the radiosity is

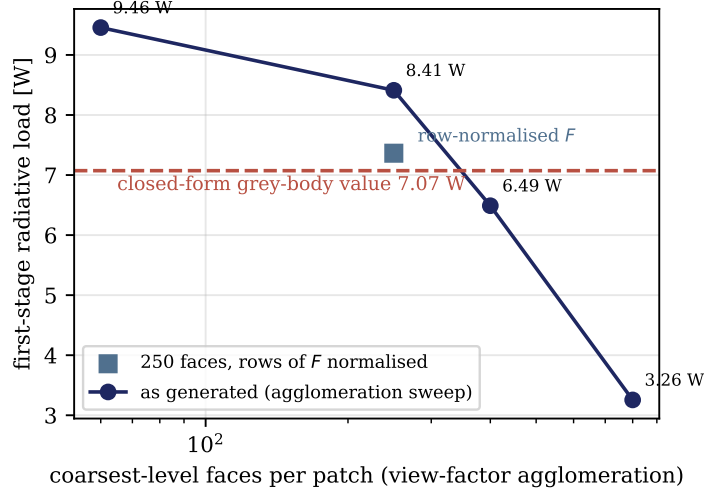


Figure 6: First-stage radiative load ($300 \rightarrow 50$ K) against the view-factor agglomeration level with the matrix as generated (circles) and with its rows normalised to unity at the production level (square), compared with the closed-form grey-body value. The monotonic collapse under refinement is a closure artefact amplified by the low emissivities.

196 reflected of order $1/\varepsilon$ times before it is absorbed, so a per-reflection leak of a
 197 few per cent compounds into a load error of tens of per cent. Normalising each
 198 row of the generated matrix to unity removes the artefact: at the production
 199 level of 250 faces the $300 \rightarrow 50$ K load becomes 7.36 W (4.1 % above the closed
 200 form), the $300 \rightarrow 77$ K load 7.34 W (4.2 %), and the two 4 K-stage loads 3.16
 201 and 17.8 mW (0.5 % above 3.15 and 17.7 mW). The remaining 4 % at the
 202 first stage is the geometric departure of the real enclosure, with its harness
 203 penetrations, from the parallel-plate idealisation. All loads below are the
 204 row-normalised values; the unnormalised production values are retained as
 205 an upper band.

206 3.2. Stage loads and mechanism split

207 At the first stage radiation dominates and the shield temperature barely
 208 matters (Table 3, Fig. 7). The radiative flux scales with $T_{\text{hot}}^4 - T_{\text{cold}}^4$ and 300^4
 209 dwarfs both 50^4 and 77^4 , so the 77 K shield intercepts 99.6 % of what the
 210 50 K shield intercepts. Harness conduction falls with the smaller span, 962
 211 against 1029 mW. The first-stage loads are 8.39 W (50 K) and 8.30 W (77 K);
 212 the emissivity band of ± 50 % moves them between 6.5 and 12.5 W but leaves

Table 3: Stage loads by mechanism (row-normalised view factors; closed-form value in parentheses), with the unnormalised production value and the $\pm 50\%$ emissivity band as upper and lower bounds.

Case	Radiation	Conduction	Total	Prod.	$\varepsilon \pm 50\%$ [W]
300 $\rightarrow$ 50 K	7.36 W (7.07)	1.029 W (1.029)	8.39 W	9.44 W	6.5–12.5
300 $\rightarrow$ 77 K	7.34 W (7.04)	0.962 W (0.962)	8.30 W	9.35 W	6.4–12.4
50 $\rightarrow$ 4 K	3.16 mW (3.15)	70.3 mW (70.3)	73.5 mW	75.6 mW	–
77 $\rightarrow$ 4 K	17.8 mW (17.7)	162.1 mW (162.1)	179.9 mW	191.8 mW	–
300 $\rightarrow$ 50 K MLI	1.81 W ^a (1.47)	1.029 W	2.84 W	–	–
300 $\rightarrow$ 77 K MLI	1.81 W ^a (1.46)	0.962 W	2.77 W	–	–

^aproduction-level matrix; the closure bias of +23 % applies as at the bare-shield level.

the 1 % difference between routes unchanged. The conservative production value of 9.4 W is used as the cold-plate duty in Section 4.

At the 4 K stage the picture inverts. Residual radiation rises by the factor $(77^4 - 4^4)/(50^4 - 4^4) = 5.6$, but from a base of 3 mW; harness conduction rises by the factor 2.31 from a base of 70 mW. The 77 K route therefore delivers 106 mW more to the 4 K plate: 92 mW of conduction and 15 mW of radiation, 86 % conductive. Through the ideal 4 K multiplier of 74 this is 7.9 W of room-temperature power, 6.8 W of it from the harness. Every stage-integrated ratio agrees with its closed form to better than 0.5 %.

The MLI-equivalent shield ($\varepsilon = 0.003$ on the first-stage surfaces) cuts the first-stage radiation to 1.81 W on either route (closed form 1.47 W with the production-level matrix bias), so the first-stage loads fall to 2.84 and 2.77 W and become conduction-heavy (36 % conduction). The 4 K-stage loads are unaffected. This variant is the one that changes the ledger verdict in Section 5.

3.3. Spatial distribution

The view-factor solution concentrates the radiative flux on the shield regions with a direct line of sight to the OVC (8.2–10.8 W m⁻² across the shield) and leaves the harness penetrations cooler (Fig. 8). Conduction is shared among the bundles in the ratio of their line counts: on the 77 $\rightarrow$ 4 K span the drive and flux families deliver 125 of the 162 mW. These two bundles are the targets for any intermediate heat sink, and the harness conduction

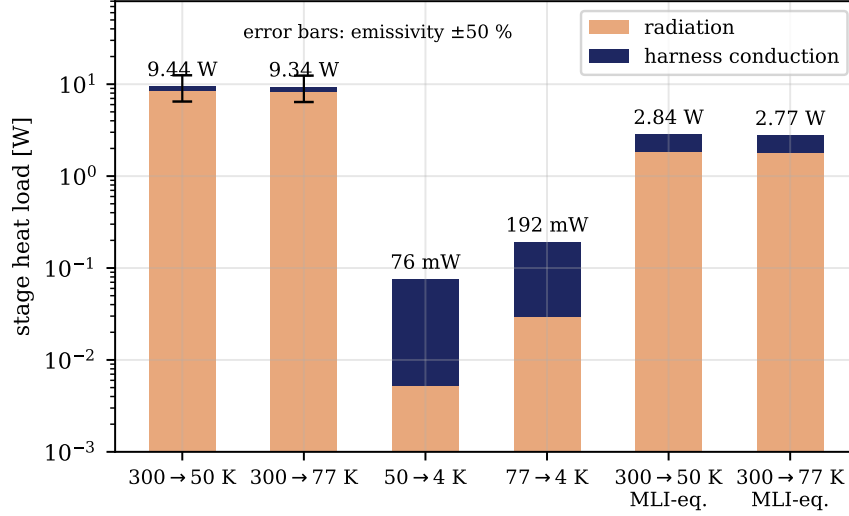


Figure 7: Stage loads for the four enclosure cases and the two MLI-equivalent variants (production view factors), decomposed by mechanism, with the $\pm 50\%$ emissivity band. The first stage is radiation-dominated and nearly identical between routes; the 4 K stage is conduction-dominated and is where the routes diverge.

integral, not the cryogen, is the lever on the 4 K-stage penalty.

4. Results Q2: helium gas versus liquid nitrogen in the cold plate

4.1. Verification

Every forced-convection run closes its energy balance, wall heat input against mass-flux-weighted enthalpy rise, to within 0.2% (Supplementary Table S7). At $Re = 500$ the helium flow is thermally developed over most of the channel ($L_t/L = 0.59$) and returns $Nu = 3.66$, 1.2% above the fully developed square-duct value of 3.61 [41]; every other laminar run is thermally developing (L_t/L up to 8.8 for nitrogen) and lies above it, as it must. The three-grid studies (Table 4) converge monotonically for both fluids: the production-mesh grid convergence index is 0.7% (Nu) and 2.5% (f) for helium and 2.2% and 1.6% for nitrogen. In the turbulent runs the first-cell y^+ on the heated wall averages 2.5–5.6 with a maximum of 9.5, within the low-Reynolds treatment of the SST model. Nitrogen’s turbulent Nusselt numbers lie 8–9% below the Gnielinski correlation [42] and helium’s 28–32% below it; the correlation is for a duct heated on all walls, and a single heated wall

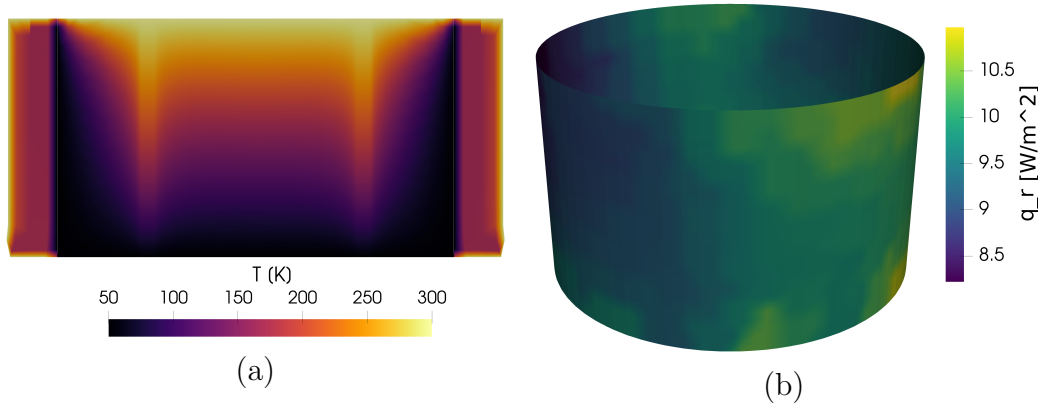


Figure 8: (a) Mid-plane temperature and (b) shield radiative-flux map of the $300 \rightarrow 50$ K case.

Table 4: Grid convergence at $Re = 2300$ (Celik et al. [43]; meshes of 86k, 292k and 984k cells, refinement ratio 1.5). The production mesh is the medium one.

Fluid	Quantity	Coarse	Medium	Fine	p	Extrapolated	GCI_{medium}
He (1 bar)	Nu	6.439	6.449	6.457	0.64	6.49	0.7 %
	f	0.0632	0.0642	0.0647	1.40	0.0655	2.5 %
LN ₂ (3 bar)	Nu	10.51	10.44	10.38	0.87	10.26	2.2 %
	f	0.0592	0.0599	0.0603	1.71	0.0607	1.6 %

penalises the low-Prandtl gas, whose thermal layer fills the duct, more than the liquid.

4.2. Operating maps

Figure 9 maps the two coolants over $Re = 500\text{--}10^4$ at the fixed 9.4 W duty. Nitrogen’s heat-transfer coefficient exceeds helium’s by a factor of 4.9–5.1 throughout the laminar range and 6.2–6.3 in the turbulent range; the laminar ratio is set by the conductivity ratio of 3.1 times the ratio of developing-flow Nusselt numbers, which favours the high-Prandtl liquid. The consequence for the plate is stark. At atmospheric pressure the helium-cooled wall reaches 98 K at $Re = 500$ and 69 K at $Re = 2300$, and falls below the 77 K nitrogen temperature only above $Re \approx 1400$ (Fig. 9d); the nominal 50 K of the helium route is reached only in the turbulent range, at $Re = 10^4$ and 65 mW of pumping power. The nitrogen-cooled wall stays between 78 and 86 K over the whole map. Its saturation margin at 3 bar is 2.2 K at $Re = 500$, 7.1 K at 2300 and 10.6 K at 10^4 ; at 1 bar the wall would exceed

the 77.2 K saturation temperature in every laminar run, so a bath-fed loop at atmospheric pressure would boil on the plate at this duty and a pressurised or subcooled loop is a requirement, not a refinement.

Pumping power separates the fluids in the opposite sense once pressure enters. At 1 bar helium needs 38 times nitrogen’s pumping power at matched Reynolds number (1.39 against 0.037 mW at $Re = 2300$), because its mass flow is 26 times smaller but its velocity 32 times larger. Raising the helium pressure at fixed Reynolds number leaves the mass flow, the Nusselt number and the wall temperature unchanged to four significant figures and reduces the pumping power as the inverse square of the density: 25.0 times at 5 bar and 324 times at 18 bar (Fig. 9c). At the 18 bar of the ITER and DTT shield loops [4, 5] helium’s pumping power is $4.3 \mu\text{W}$, 8.5 times below nitrogen’s; the parallel-path gaseous-helium shields of LHD operate in the same regime [6].

4.3. Wall-temperature distributions

Figure 10 gives the heated-wall centreline temperature and the cross-channel profile at mid-length, the quantitative counterpart of the rendered fields of the earlier comparison. Along the wall the temperature rise grows as $x^{1/3}$ in the developing region and then linearly; at the outlet the helium wall sits 24 K above its inlet at $Re = 2300$ and 53 K at $Re = 500$, nitrogen 4.2 and 8.7 K. Across the channel the helium thermal layer at $Re = 500$ fills the duct, whereas nitrogen’s occupies the near-wall 30 % even at $Re = 500$. This is the geometric origin of the heat-transfer ratio and of the plate temperatures in Fig. 9d.

4.4. Ranking on three bases

Table 5 collects the design point and Fig. 11 the trade-off between wall superheat and pumping power. The ranking depends on the basis. At matched Reynolds number nitrogen wins heat transfer by a factor of 5 and pumping power by 38; the effectiveness–NTU pair favours helium (0.11 against 0.06), a capacity-rate artefact of the gas’s small $\dot{m}c_p$ that says nothing about how cold the plate is. At matched mass flow nitrogen’s Reynolds number falls to about 90 and its heat-transfer coefficient to the fully developed limit of about $53 \text{ W m}^{-2}\text{K}^{-1}$, still 1.8 times helium’s at $Re = 2300$. At matched pumping power, the basis a plant designer pays for, nitrogen at the pumping power of 1 bar helium ($Re = 2300$) holds a superheat of 0.9 K against helium’s 20.8 K, a factor of 23; at the pumping power of 18 bar helium the factor is 3.5 (5.9 against 20.8 K). In the turbulent range the curves in Fig. 11 are parallel: a

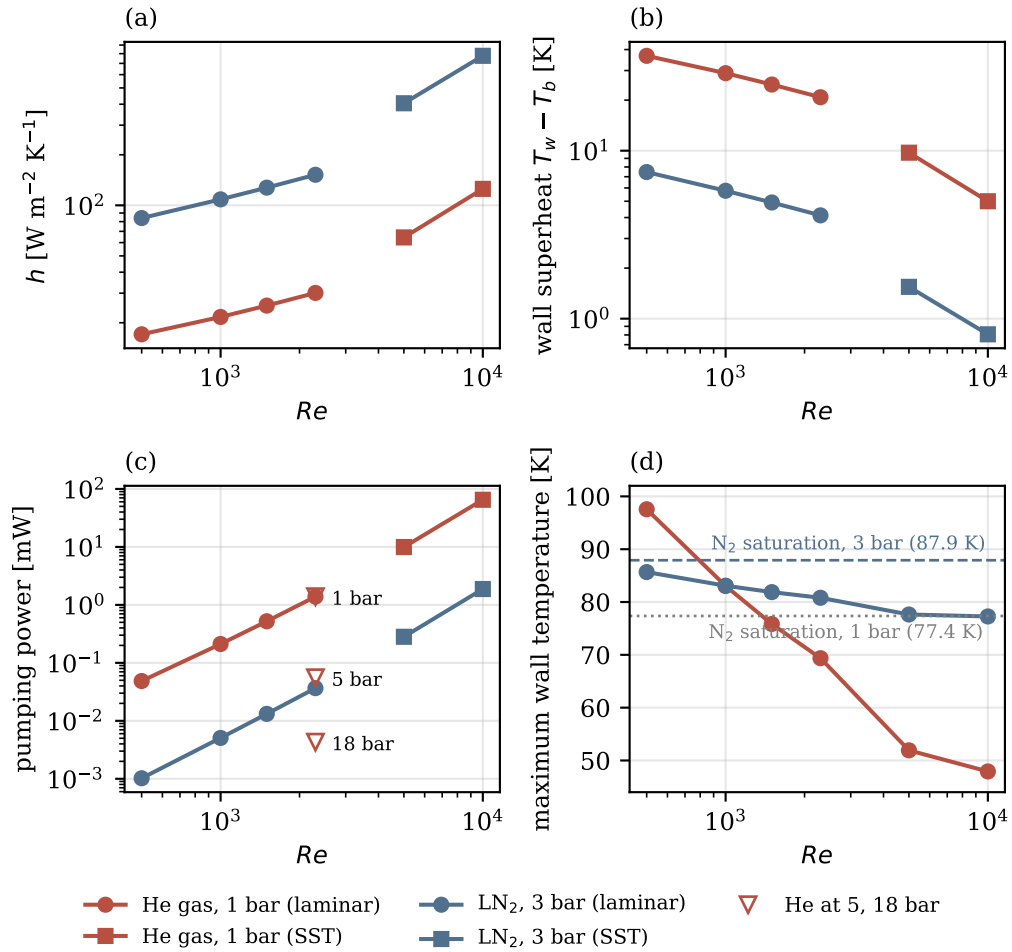


Figure 9: Operating maps at 9.4 W duty: (a) heat-transfer coefficient, (b) wall superheat, (c) pumping power with the helium pressure series at $Re = 2300$, and (d) maximum wall temperature with the nitrogen saturation temperatures at 1 and 3 bar. Circles: laminar; squares: $k-\omega$ SST.

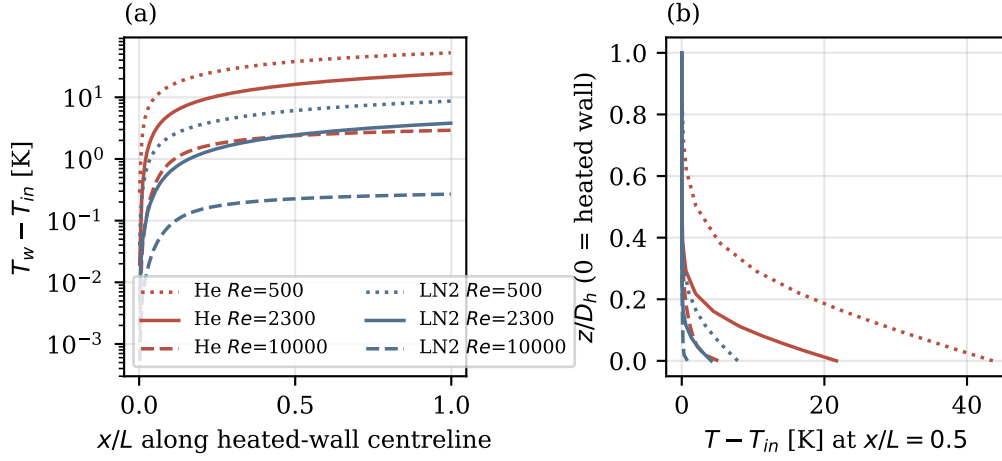


Figure 10: (a) Heated-wall centreline temperature rise above the inlet and (b) cross-channel temperature profile at $x/L = 0.5$ for both fluids at $Re = 500$, 2300 and 10^4 .

302 given superheat costs helium at 1 bar roughly 30 times the pumping power it
 303 costs nitrogen, and the 18 bar helium point sits inside the nitrogen envelope
 304 on pumping power alone but outside it on superheat. No single number ranks
 305 the coolants; the operating point and the pressure do.

306 4.5. Buoyancy sensitivity

307 The buoyancy parameter Gr/Re^2 of the forced-convection runs is below
 308 0.6 for helium and above unity for nitrogen in the laminar range (40 at
 309 $Re = 500$, 1.05 at 2300), so a real plate in which gravity acts along the heated
 310 wall would carry a mixed-convection secondary flow in nitrogen. The three
 311 runs with gravity on confirm this and bound its effect. Helium converged
 312 and gained 27% in Nu at $Re = 2300$ and 118% at $Re = 500$. Nitrogen did
 313 not reach a steady state at either Reynolds number: after 10 000 iterations
 314 the temperature residual oscillated at 10^{-3} and the energy balance remained
 315 open by 11% and 72%, the signature of an unsteady buoyant flow that a
 316 steady solver cannot represent. The forced-convection nitrogen results are
 317 therefore lower bounds on its heat transfer, and the ranking of Section 4.4 is
 318 conservative toward helium.

Table 5: Design point, $Re = 2300$, 9.4 W, forced convection. He at 1 bar (18 bar in parentheses where pressure matters); LN₂ at 3 bar. GCI on Nu and h : 0.7 % (He), 2.2 % (LN₂).

Quantity	He, 50 K	LN ₂ , 77 K	LN ₂ /He
Mass flow $\dot{m}$ [g s ⁻¹]	0.731	18.8	25.7
Volumetric flow [cm ³ s ⁻¹]	702 (39)	23.3	0.033
Mean velocity U [m s ⁻¹]	1.52 (0.084)	0.047	0.031
Inlet / outlet temperature [K]	45.0 / 47.5	77.0 / 77.2	–
Pressure drop Δp [Pa]	1.98 (0.11)	1.57	0.80
Pumping power $\dot{W}_p$ [μ W]	1388 (4.3)	36.6	0.026 (8.5)
Wall temperature, mean / max [K]	67.1 / 69.4	81.2 / 80.8 ^a	–
Wall superheat $T_w - T_b$ [K]	20.8	4.12	0.20
Heat-transfer coefficient h [W m ⁻² K ⁻¹]	30.1	152	5.05
Nusselt number Nu	6.45	10.4	1.62
Darcy friction factor f	0.064	0.060	0.93
NTU / effectiveness ε	0.119 / 0.112	0.060 / 0.058	0.50 / 0.52
Mouromtseff number	567 (5720)	4.6×10^4	81 (8)
Buoyancy parameter Gr/Re^2	0.02	1.05	–
Saturation margin $T_{sat} - T_{w,max}$ [K]	–	7.1	–
Energy-balance closure	0.06 %	0.001 %	–

^acentreline maximum; the area mean includes the corner regions.

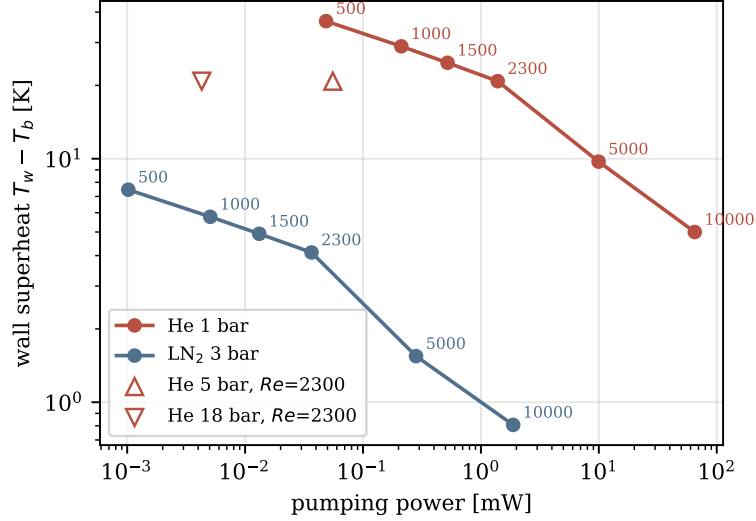


Figure 11: Wall superheat against pumping power for both fluids over the Reynolds sweep (labels: Re), with the helium pressure series at $Re = 2300$.

319 5. Results Q3: the work ledger

320 Table 6 and Fig. 12 price the two routes with Eq. (7). With ideal multi-
 321 pliers and the bare shield the nitrogen route is cheaper: 37.3 against 47.4 W,
 322 a margin of 21 %, because intercepting 8.3 W at a multiplier of 2.9 instead
 323 of 5.0 saves 18 W while the 4 K-stage penalty costs 7.9 W. The emissivity
 324 band (31.8–49.2 W against 37.8–68.0 W) does not close the gap. With the
 325 MLI-equivalent shield the first-stage load shrinks to 2.8 W, the nitrogen sav-
 326 ing to 6.2 W, and the verdict flips: helium 19.6 W against nitrogen 21.3 W.
 327 The ideal ledger therefore favours nitrogen where the first-stage load is large
 328 and helium where insulation has removed it, which is the regime of the MLI-
 329 wrapped shields surveyed by Shu et al. [3] and of the single-layer intercepts
 330 of Singh et al. [44].

331 With hardware efficiencies the verdict is not close in either case. The
 332 4 K stage at 0.8–1.5 % of Carnot turns the 106 mW penalty into 520–980 W,
 333 more than the whole first-stage bill of either route, and the helium route
 334 costs 840 W (640–1100) against 1290 W (950–1740) for nitrogen with the
 335 bare shield and 610 against 1240 W with MLI. Written as a break-even, the
 336 nitrogen route is cheaper only if

$$\frac{\eta_2}{\eta_1} \gtrsim \frac{74 \Delta \dot{Q}_2}{5.0 \dot{Q}_{1,\text{He}} - 2.9 \dot{Q}_{1,\text{N}_2} \eta_1 / \eta_{\text{N}_2}}, \quad (8)$$

337 which with the present loads, $\eta_1 = 0.12$ and $\eta_{\text{N}_2} = 0.35$ requires a 4 K-
 338 stage efficiency of 2.8 % of Carnot for the bare shield and 8 % for the MLI
 339 shield, two to eight times what commercial coolers achieve [30]. The lever
 340 is $\Delta \dot{Q}_2$: because 86 % of it is harness conduction, a harness engineered for a
 341 threefold lower $77 \rightarrow 4$ K conduction integral, by heat-sinking the drive and
 342 flux bundles or by lower-conductivity outer conductors, brings the bare-shield
 343 break-even to 1.2 % of Carnot, within the range of real hardware.

344 6. Discussion

345 *The penalty is a harness property..* The passive cost of the warmer shield is
 346 located, sized and attributed: 106 mW at the 4 K stage, 86 % of it conduction
 347 along a harness whose drive and flux families carry three quarters. It is a
 348 property of the wiring, not of the cryogen, and Eq. (8) shows that the wiring
 349 is the only variable that can move the break-even into the range of real
 350 hardware. Lumped budgets give the same total; what the resolved model

Table 6: Work ledger: stage loads (row-normalised view factors), ideal and hardware-based room-temperature power. Ranges: emissivity $\pm 50\%$ (ideal) and stage-efficiency ranges of Supplementary Table S9 (real).

Shield	Route	$\dot{Q}_1$ [W]	$\dot{Q}_2$ [mW]	$\dot{W}_{\text{ideal}}$ [W]	$\dot{W}_{\text{real}}$ [W]
bare	He, 50 K	8.39	73.5	47.4 (37.8–68.0)	840 (640–1100)
bare	LN ₂ , 77 K	8.30	179.9	37.3 (31.8–49.2)	1290 (950–1740)
MLI-eq.	He, 50 K	2.84	73.5	19.6	610 (460–820)
MLI-eq.	LN ₂ , 77 K	2.77	179.9	21.3	1240 (910–1690)

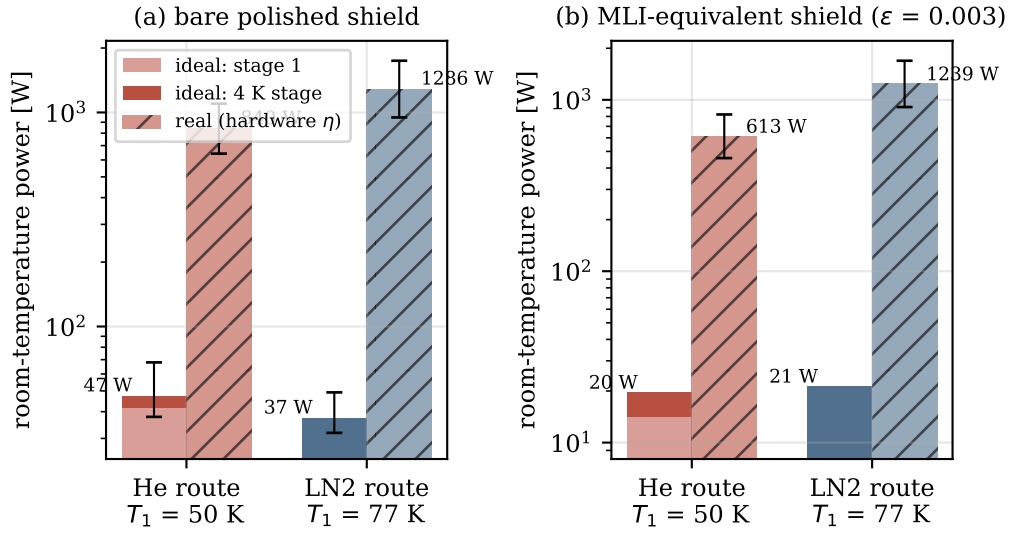


Figure 12: Room-temperature power of the two routes for (a) the bare polished shield and (b) the MLI-equivalent shield: ideal Carnot multipliers split by stage (solid) and hardware efficiencies (hatched), with emissivity and efficiency ranges as error bars.

351 adds is the attribution to bundles that can be heat-sunk, and the confirmation
 352 that the residual radiation, whatever the shield temperature, is a minor term.
 353 This is the design steer that Li et al.’s optimisation of intercepts on supports
 354 and in insulation [11] cannot give for a wired cryostat.

355 *The helium plate floats..* A cold plate cooled by circulated helium gas at
 356 atmospheric pressure does not sit at the gas temperature. At the 9.4 W duty
 357 it runs 21 K above the bulk at $Re = 2300$ and reaches nitrogen temperature
 358 at $Re \approx 1400$; the nominal 50 K is recovered only in turbulent flow at tens
 359 of milliwatts of pumping power, or by pressurising the loop, which leaves the
 360 wall temperature untouched but makes the pumping power negligible. This
 361 is a statement about the circulated-gas architecture and the channel sizing,
 362 not about the conductively coupled PTR first stage of Fig. 2a, whose plate
 363 temperature is set by the cooler’s capacity curve. It does mean that the
 364 50/77 K distinction of the passive comparison is only realised on the helium
 365 route with an adequate flow or pressure, and Section 4.2 gives the conditions.

366 *The basis of comparison decides the ranking..* Matched Reynolds number,
 367 matched mass flow and matched pumping power rank the fluids differently
 368 in magnitude and, for pumping power at 18 bar, in sign. The effectiveness–
 369 NTU pair inverts the engineering conclusion because it rewards a small ca-
 370 pacity rate. A cryogenic coolant comparison must therefore state its basis
 371 and its operating point, as Gačnik et al. found necessary for nitrogen-cooled
 372 terminals [29]; the operating maps of Fig. 9 are offered as the form such a
 373 comparison should take.

374 *Ideal and real ledgers disagree, and the disagreement is informative..* The
 375 ideal ledger prefers nitrogen for a bare shield and helium for an insulated
 376 one; the hardware ledger prefers helium in both cases by 35–50 %, because
 377 the 4 K stage of a commercial cooler converts every milliwatt of penalty into
 378 ten watts at the wall. The nitrogen route retains what Li et al. identified
 379 three decades ago [11]: a constant shield temperature, stored reserve, and
 380 independence from the cryocooler, which now also means freedom from its
 381 vibration [45]. Those are operational arguments; the thermodynamic one is
 382 conditional on the harness.

383 *Methodological findings..* Two travel beyond this geometry. First, view-
 384 factor matrices generated for low-emissivity enclosures must be checked for
 385 closure and, if the enclosure is closed, row-normalised; the required closure

accuracy is of order the emissivity, and agglomeration refinement does not supply it. The benchmark practice of Končar et al. [25] should be extended by this test. Second, the mixing-cup temperature on a graded mesh must be mass-flux weighted without an additional area factor; the corrected accounting closed the energy balance of every run from a few per cent to below 0.2 %.

Limitations.. The enclosure imposes fixed temperatures on its plates and shield and neglects contact resistance at the harness ends; a finitely conductive shield would carry the radiative-flux map of Fig. 8 into a temperature map, which is the next step. The cold plate is a single-wall-heated smooth duct with constant properties except for the helium density and the nitrogen buoyancy term; conjugate conduction in the plate, which would spread the wall temperature, is not resolved. Nitrogen is single phase by verified margin at 3 bar but not at 1 bar. The turbulent runs rely on the SST model at $y^+ \leq 9.5$ and were not compared with experiment. The hardware efficiencies are ranges from published data, and the allocation of a two-stage cooler's input between its stages is by Carnot work; a measured capacity map of a specific machine would replace the ranges. Buoyancy in nitrogen was shown to be unsteady and was excluded from the ranking rather than resolved; a transient calculation in a specified plate orientation would quantify it.

7. Conclusions

1. A 77 K first shield costs nothing at the first stage relative to a 50 K shield: it intercepts 99.6 % of the radiation and 6.5 % less harness conduction, 8.3 against 8.4 W for the enclosure studied. The penalty appears at the 4 K stage, 106 mW, and is 86 % harness conduction concentrated in the drive and flux bundles; it is worth 7.9 W of ideal room-temperature power and 520–980 W with real 4 K coolers.
2. View-factor radiation in low-emissivity enclosures is controlled by matrix closure: the generated matrix gave loads from 9.5 to 3.3 W under agglomeration refinement against a 7.07 W closed form, and row normalisation recovered the closed form to 4 % at the first stage and 0.5 % at the 4 K stage.
3. As a first-shield coolant at 9.4 W, nitrogen at 3 bar gives 5 times helium's heat-transfer coefficient in laminar flow and 6 times in turbulent flow, and holds the plate within 4 K of its inlet at $Re = 2300$ against

helium’s 21 K; helium at 1 bar lets the plate float above 77 K below $Re \approx 1400$ and needs 38 times the pumping power, but at 18 bar its pumping power falls below nitrogen’s by a factor of 8.5 with heat transfer unchanged. The ranking depends on the basis of comparison and the operating point.

4. The ideal ledger favours nitrogen by 21 % for a bare shield and helium by 8 % for an MLI shield; with hardware efficiencies helium is cheaper by 35–50 % in both cases. The nitrogen route becomes competitive with real hardware only if the $77 \rightarrow 4$ K harness conduction is cut by about a factor of three.
5. The corrected helium properties, the forced-convection basis, the mass-flux-weighted mixing-cup temperature and the row-normalised view factors together change the earlier quantitative picture of this comparison; the qualitative conclusion, that the shield choice and the harness design are not separable, is unchanged and now rests on verified numbers.

CRedit authorship contribution statement

Imerson Joao: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

All OpenFOAM cases (enclosure baseline and variants, cold-plate sweep, grid studies), the post-processing and figure scripts, and the tabulated results are available at github.com/Imerson/Pre-cooling-a-superconducting-qubit-dilution-refrigerator (release v2); an archived copy will be deposited on Zenodo on acceptance.

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453 Declaration of generative AI in scientific writing

454 During the preparation of this work the author used Claude (Anthropic)
455 to assist with editing, revision and consistency checking of the manuscript
456 and with scripting the simulation campaign. The author reviewed and edited
457 all content and takes full responsibility for the published article.

458 References

- 459 [1] J. G. Weisend II, Cryostat Design: Case Studies, Principles and Engi-
460 neering, Springer, Cham, 2016. doi:10.1007/978-3-319-31150-0.
461 URL <https://doi.org/10.1007/978-3-319-31150-0>
- 462 [2] R. F. Barron, G. F. Nellis, Cryogenic Heat Transfer, 2nd Edition, CRC
463 Press, Boca Raton, FL, 2016.
- 464 [3] Q.-S. Shu, J. Demko, J. Fesmire, R. Duckworth, Design, configuration,
465 and thermal optimization of advanced cryostats, IOP Conference Series:
466 Materials Science and Engineering 1301 (2024) 012041. doi:10.1088/
467 1757-899X/1301/1/012041.
- 468 [4] K. Nam, W. Chung, C. H. Noh, D. K. Kang, K.-O. Kang, H. J. Ahn,
469 et al., Thermal analysis on detailed 3D models of ITER thermal shield,
470 Fusion Engineering and Design 89 (2014) 1843–1847. doi:10.1016/j.
471 fusengdes.2014.02.060.
- 472 [5] G. Barone, S. Roccella, E. Martelli, E. Visca, DTT thermal shield: pre-
473 liminary thermal analysis, Fusion Engineering and Design 158 (2020)
474 111725. doi:10.1016/j.fusengdes.2020.111725.
- 475 [6] S. Imagawa, Performance of thermal shields of LHD cryostat cooled
476 by gaseous helium with parallel paths, in: Advances in Cryogenic
477 Engineering: Proceedings of the Cryogenic Engineering Conference –
478 CEC, Vol. 613 of AIP Conference Proceedings, 2002, pp. 1475–1482.
479 doi:10.1063/1.1472180.
- 480 [7] P. Lebrun, Cryogenics for the Large Hadron Collider, IEEE Transactions
481 on Applied Superconductivity 10 (1) (2000) 1500–1506. doi:10.1109/
482 77.828526.

- 483 [8] S. Claudet, P. Lebrun, L. Tavian, U. Wagner, Exergy analysis of the
484 cryogenic helium distribution system for the Large Hadron Collider
485 (LHC), in: *Advances in Cryogenic Engineering: Transactions of the*
486 *Cryogenic Engineering Conference – CEC*, Vol. 1218 of AIP Conference
487 *Proceedings*, 2010, pp. 1267–1274. doi:10.1063/1.3422294.
- 488 [9] K. Pieterman, J. A. M. Dam, H. Postma, Low temperature closed cool-
489 ing system for mri magnets, in: R. W. Fast (Ed.), *Advances in Cryo-*
490 *genic Engineering*, Vol. 31, Springer, Boston, MA, 1986, pp. 533–541.
491 doi:10.1007/978-1-4613-2213-9_61.
- 492 [10] Y. Iwasa, *Case Studies in Superconducting Magnets: Design and Oper-*
493 *ational Issues*, 2nd Edition, Springer, New York, 2009. doi:10.1007/
494 b112047.
- 495 [11] Q. Li, X. Li, G. E. McIntosh, R. W. Boom, Minimization of total refriger-
496 ation power of liquid neon and nitrogen cooled intercepts for SMES mag-
497 nets, in: *Advances in Cryogenic Engineering*, Vol. 35, Springer, 1990,
498 pp. 833–840. doi:10.1007/978-1-4613-0639-9_100.
- 499 [12] K. Uhlig, “dry” dilution refrigerator with pulse-tube precooling, *Cryo-*
500 *genics* 44 (1) (2004) 53–57. doi:10.1016/j.cryogenics.2003.07.007.
- 501 [13] A. T. A. M. de Waele, Basic operation of cryocoolers and related thermal
502 machines, *Journal of Low Temperature Physics* 164 (5–6) (2011) 179–
503 236. doi:10.1007/s10909-011-0373-x.
504 URL <https://doi.org/10.1007/s10909-011-0373-x>
- 505 [14] S. Krinner, S. Storz, P. Kurpiers, P. Magnard, J. Heinsoo, R. Keller,
506 F. Lüthi, C. Eichler, A. Wallraff, Engineering cryogenic setups for 100-
507 qubit scale superconducting circuit systems, *EPJ Quantum Technology*
508 6 (2), received: 21 November 2018; Accepted: 04 April 2019; Published:
509 28 May 2019 (2019). doi:10.1140/epjqt/s40507-019-0072-0.
510 URL <https://doi.org/10.1140/epjqt/s40507-019-0072-0>
- 511 [15] P. Gumann, J. Chow, Ibm scientists cool down the world’s largest
512 quantum-ready cryogenic concept system (2022).
513 URL [https://www.ibm.com/quantum/blog/](https://www.ibm.com/quantum/blog/goldeneye-cryogenic-concept-system)
514 [goldeneye-cryogenic-concept-system](https://www.ibm.com/quantum/blog/goldeneye-cryogenic-concept-system)

- [16] M. Hollister, R. Dhuley, G. Tatkowski, A large millikelvin platform at fermilab for quantum computing applications, arXiv preprint arXiv:2108.10816 (2021). **arXiv:2108.10816**.
- [17] A. Auffèves, Quantum technologies need a quantum energy initiative, PRX Quantum 3 (2022) 020101. doi:10.1103/PRXQuantum.3.020101. URL <https://link.aps.org/doi/10.1103/PRXQuantum.3.020101>
- [18] M. De Michielis, E. Ferraro, Energy and power scaling in quantum computers based on rotated surface codes with silicon flip-flop qubits, EPJ Quantum Technology 12 (64) (2025). doi:10.1140/epjqt/s40507-025-00351-4. URL <https://doi.org/10.1140/epjqt/s40507-025-00351-4>
- [19] M. J. Martin, C. Hughes, G. Moreno, E. B. Jones, D. Sickinger, S. Narumanchi, R. Grout, Energy use in quantum data centers: Scaling the impact of computer architecture, qubit performance, size, and thermal parameters, IEEE Transactions on Sustainable Computing 7 (4) (2022) 864–874. doi:10.1109/TSUSC.2021.3050774. URL <https://doi.org/10.1109/TSUSC.2021.3050774>
- [20] C. Miles, R. Trepanier, et al., Cryostat optimization through multiple stage thermal shields, in: Proceedings of the 2008 Applied Superconductivity Conference / ICEC, 2008, superconducting transmission-line cryostats; multiple-shield refrigeration-power reduction.
- [21] C. Groth, A. Chiappa, R. Guarino, An analytical thermal model for the optimization of EU DEMO feeder thermal shields, Fusion Engineering and Design 207 (2024) 114650. doi:10.1016/j.fusengdes.2024.114650.
- [22] X. Zhang, M. Xu, Z. Chen, X. Yang, X. Yang, T. Zhao, R. Ye, X. Bian, Thermal performance research on the zero liquid helium consumption cryostat for a superconducting undulator, Energy 308 (2024) 132868. doi:10.1016/j.energy.2024.132868.
- [23] N. Raicu, T. Hogan, X. Wu, M. Vahidpour, D. Snow, M. Hollister, M. Field, Cryogenic thermal modeling of microwave high density signaling, EPJ Quantum Technology 12 (2025) 124. doi:10.1140/epjqt/s40507-025-00427-1.

- [24] R. Snodgrass, J. Ullom, S. Backhaus, Optimal absorption of distributed and conductive heat loads with cryocooler regenerators, IOP Conference Series: Materials Science and Engineering 1240 (2022) 012131. doi:10.1088/1757-899X/1240/1/012131.
- [25] B. Končar, L. Kostanjšek, M. Venčelj, Use of CFD codes for calculation of radiation heat transfer: from validation to application, in: Heat Transfer – Models, Methods and Applications, IntechOpen, 2018. doi:10.5772/intechopen.74420.
- [26] P. P. P. M. Lerou, T. T. Veenstra, J. F. Burger, H. J. M. ter Brake, H. Rogalla, Optimization of counterflow heat exchanger geometry through minimization of entropy generation, Cryogenics 45 (10–11) (2005) 659–669. doi:10.1016/j.cryogenics.2005.08.002.
- [27] Ø. Wilhelmsen, D. Berstad, A. Aasen, P. Neksa, G. Skaugen, Reducing the exergy destruction in the cryogenic heat exchangers of hydrogen liquefaction processes, International Journal of Hydrogen Energy 43 (10) (2018) 5033–5047. doi:10.1016/j.ijhydene.2018.01.094.
- [28] A. Bejan, Entropy Generation Minimization: The Method of Thermodynamic Optimization of Finite-Size Systems and Finite-Time Processes, CRC Press, Boca Raton, FL, 1996.
- [29] D. Gačnik, M. ter Brake, M. Dhallé, Thermal modeling of LN₂-cooled terminals with integrated heat exchangers for superconducting applications, Applied Thermal Engineering 279 (2025) 127491. doi:10.1016/j.applthermaleng.2025.127491.
- [30] R. Radebaugh, Cryocoolers: the state of the art and recent developments, Journal of Physics: Condensed Matter 21 (16) (2009) 164219. doi:10.1088/0953-8984/21/16/164219.
- [31] A. Karabuğa, R. Selbaş, A. Kabul, Energy and exergy analyses of nitrogen liquefaction process, in: I. Dincer, C. O. Colpan, O. Kizilkan (Eds.), Exergetic, Energetic and Environmental Dimensions, Academic Press, 2018, pp. 627–639. doi:10.1016/B978-0-12-813734-5.00036-6.
- [32] I. H. Bell, J. Wronski, S. Quoilin, V. Lemort, Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermo-

- 580 physical property library CoolProp, Industrial & Engineering Chemistry
581 Research 53 (6) (2014) 2498–2508. doi:10.1021/ie4033999.
- 582 [33] Bluefors Oy, PT415 pulse tube cryocooler: technical specifica-
583 tions, [https://bluefors.com/products/pulse-tube-cryocoolers/](https://bluefors.com/products/pulse-tube-cryocoolers/pt415-pulse-tube-cryocooler/)
584 [pt415-pulse-tube-cryocooler/](https://bluefors.com/products/pulse-tube-cryocoolers/pt415-pulse-tube-cryocooler/), 1.5 W at 4.2 K with 40 W at 45
585 K; compressor input 10.7 kW (60 Hz) / 9.2 kW (50 Hz); accessed 27
586 September 2026 (2026).
- 587 [34] OpenCFD Ltd., OpenFOAM: The Open Source CFD Toolbox, User
588 Guide, Version v2412, OpenCFD Ltd. (ESI Group) (2024).
589 URL <https://www.openfoam.com/documentation/user-guide>
- 590 [35] T. L. Bergman, A. S. Lavine, F. P. Incropera, D. P. DeWitt, Funda-
591 mentals of Heat and Mass Transfer, 8th Edition, John Wiley & Sons,
592 Hoboken, NJ, 2017.
- 593 [36] F. R. Menter, Two-equation eddy-viscosity turbulence models for en-
594 gineering applications, AIAA Journal 32 (8) (1994) 1598–1605. doi:
595 10.2514/3.12149.
- 596 [37] J. W. Ekin, Experimental Techniques for Low-Temperature Measure-
597 ments: Cryostat Design, Material Properties and Superconductor
598 Critical-Current Testing, Oxford University Press, Oxford, 2006.
- 599 [38] Q. S. Shu, R. W. Fast, H. L. Hart, Heat flux from 277 to 77 K through
600 a few layers of multilayer insulation, Cryogenics 26 (12) (1986) 671–677.
601 doi:10.1016/0011-2275(86)90167-0.
- 602 [39] W. M. Kays, A. L. London, Compact Heat Exchangers, 3rd Edition,
603 McGraw-Hill, New York, 1984.
- 604 [40] I. E. Mouromtseff, Water and forced-air cooling of vacuum tubes, Pro-
605 ceedings of the IRE 30 (4) (1942) 190–205. doi:10.1109/JRPROC.1942.
606 234654.
607 URL <https://doi.org/10.1109/JRPROC.1942.234654>
- 608 [41] R. K. Shah, A. L. London, Laminar Flow Forced Convection in Ducts,
609 Advances in Heat Transfer, Supplement 1, Academic Press, New York,
610 1978.

- 611 [42] V. Gnielinski, New equations for heat and mass transfer in turbulent
612 pipe and channel flow, *International Chemical Engineering* 16 (2) (1976)
613 359–368.
- 614 [43] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman, P. E.
615 Raad, Procedure for estimation and reporting of uncertainty due to
616 discretization in CFD applications, *Journal of Fluids Engineering* 130 (7)
617 (2008) 078001. doi:10.1115/1.2960953.
618 URL <https://doi.org/10.1115/1.2960953>
- 619 [44] M. K. Singh, V. Singh, K. Saraswat, D. Singh, Potential of ther-
620 mal protection improvements in large-scale cryostat development, *Inter-
621 national Journal of Heat and Mass Transfer* 236 (2025) 126254.
622 doi:10.1016/j.ijheatmasstransfer.2024.126254.
- 623 [45] T. Tomaru, T. Suzuki, T. Haruyama, T. Shintomi, A. Yamamoto,
624 T. Koyama, R. Li, Vibration analysis of cryocoolers, *Cryogenics* 44 (5)
625 (2004) 309–317. doi:10.1016/j.cryogenics.2003.12.006.
626 URL <https://doi.org/10.1016/j.cryogenics.2003.12.006>